Right-angled triangles

1

a AB

b BC

2

a AC

b BC

c BC

d AC

e $\angle ACB$

3

a i $\frac{4}{3}$ ii $\frac{4}{5}$ iii $\frac{3}{5}$

b i $\frac{15}{8}$ ii $\frac{15}{17}$ iii $\frac{8}{17}$

c i $\frac{21}{20}$ ii $\frac{21}{29}$ iii $\frac{20}{29}$

d i $\frac{12}{5}$ ii $\frac{12}{13}$ iii $\frac{5}{13}$

e i $\frac{24}{7}$ ii $\frac{24}{25}$ iii $\frac{7}{25}$

f i $\frac{28}{45}$ ii $\frac{28}{53}$ iii $\frac{45}{53}$

Trigonometric ratios

4

a i $\tan\theta = \frac{AC}{BC}$ ii $\sin\alpha = \frac{BC}{AB}$

b i $\sin\theta = \frac{AC}{AB}$ ii $\tan\alpha = \frac{BC}{AC}$

c i $\cos\theta = \frac{BC}{AB}$ ii $\cos\alpha = \frac{AC}{AB}$

d i $\tan\theta = \frac{8}{15}$ ii $\sin\alpha = \frac{15}{17}$

e i $\cos\theta = \frac{15}{17}$ ii $\tan\alpha = \frac{15}{8}$

f i $\sin\theta = \frac{24}{25}$ ii $\cos\alpha = \frac{24}{25}$

5

a $x = 10$

b $\frac{4}{5}$

c $\frac{3}{5}$

6

a $x = 4$

b $\frac{4}{3}$

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Unknown sides

7 $b = 8$

8 $d = 3084$

9 $b = 19.8$

10

a $f = 4.08$ **b** $f = 6.12$ **c** $g = 9.46$ **d** $a = 11.59$

Unknown angles

11 $\frac{\sqrt{15}}{7}$

12

a $x = 67^\circ$ **b** $x = 74^\circ$

13 $A = 66.4^\circ$

14 $\tan\theta = \frac{24}{7}$

Applications

15

a $h = 21$ **b** $f = 22$ **c** $h = 98, x = 28$ **d** $x = 34$

e $x = 12$

16 $d = 1.48$

17 $1.96\ \text{m}$

18 $y = 58.02$

19

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a $x = 21.85$

b $h = 23.95 \text{ m}$

20 $\theta = 9.88^\circ$

21 $y = 54.90^\circ$

22 $\tan\theta = \frac{4}{3}$

23 $h = 4.4 \text{ m}$

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